Predicting Water Well Functionality in Tanzania

This presentation introduces a crucial project aimed at improving water access in rural Tanzania. We'll explore how data-driven insights can transform reactive maintenance into a proactive strategy, ensuring communities have consistent access to clean water.

The Challenge & The Solution

Why Proactive Maintenance Matters

Reactive is Inefficient

Traditionally, wells are repaired only after complete failure, leading to prolonged water shortages and increased costs.

Proactive Saves Lives

Our predictive model anticipates well failures, allowing for timely intervention and uninterrupted water supply.

Data-Driven Decisions

This approach shifts from guesswork to insights, optimizing resource allocation and reducing downtime for communities.

Guiding Our Path to Success

01

Primary Objective

Accurately identify wells that are functional but require repair, enabling timely maintenance before complete breakdown.

02

Secondary Objective

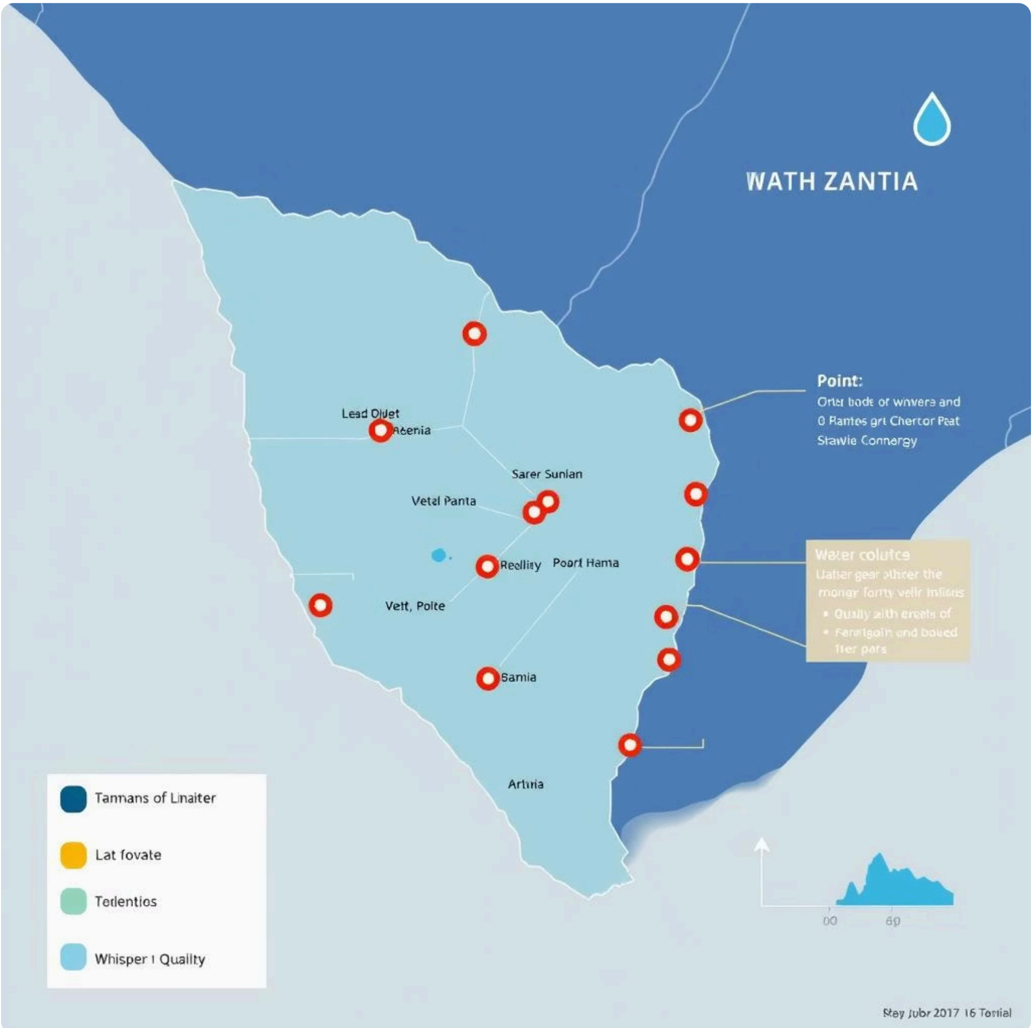
Uncover patterns in non-functional wells to inform better design and strategic placement of new water points.

03

Success Metrics

Our model's performance will be rigorously evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC.

Exploring the Data: Tanzania Water Points



Our dataset, from the Tanzania Ministry of Water, contains thousands of records crucial for understanding water well status.

Key Features:

- **Target Variable:** status_group (functional, functional but needs repair, non-functional)
- **Location:** longitude, latitude, basin, region
- **Technical:** amount_tsh, gps_height, construction_year
- **Management:** funder, installer, management, scheme_management
- **Water Source:** source_type, water_quality, quantity

Preparing the Data for Modeling

Data Integrity Check

Thoroughly inspected for missing values and identified columns needing imputation or special handling.

Addressing Class Imbalance

Recognized the critical challenge of a minority 'needs repair' class, which could skew model performance.

SMOTE for Balance

Applied Synthetic Minority Over-sampling Technique (SMOTE) to achieve a balanced representation across all classes.

This step is crucial to ensure our model doesn't overlook wells that are functional but on the brink of failure.

Training and Tuning the Model

Model Selection & Comparison

We trained and rigorously compared two robust machine learning models:

- **Random Forest:** An ensemble method known for its accuracy and ability to handle complex datasets.
- **XGBoost:** A powerful gradient boosting framework, often a top performer in classification tasks.

Hyperparameter Tuning

Using **RandomizedSearchCV**, we optimized key hyperparameters:

- **n_estimators:** Number of boosting rounds or trees.
- **max_depth:** Maximum depth of a tree.
- **learning_rate:** Step size shrinkage to prevent overfitting.

The tuned **XGBoost** model demonstrated superior performance and was selected as our final classifier.

Analyzing the Model's Performance

Our XGBoost model achieved a respectable overall accuracy, but a deeper look at the classification report reveals critical insights.

Functional	83%	84%	87%
Functional Needs Repair	28%	48%	26%
Non-Functional	74%	78%	73%

Overall Accuracy: **79%**

⚠ The model excels at identifying truly functional wells, but severely struggles with the crucial "needs repair" category.

Recommendations for Improvement

Feature Engineering

Explore and create new features that could better highlight subtle differences in wells needing repair.

Alternative Imbalance Handling

Consider **class weighting** or other advanced techniques instead of SMOTE, which might introduce noise.

Expand Model Search

Broaden the hyperparameter search space or explore algorithms like **LightGBM** for better imbalance handling.

These steps will be critical in improving the model's ability to identify wells requiring preventative maintenance.

Looking Ahead: Enhancing Our Predictive Power

Project Summary

Successfully developed a predictive model for well functionality, with strong performance in identifying functional and non-functional wells.

Main Challenge

The low recall for the 'functional needs repair' class is a significant limitation, hindering truly proactive maintenance.

The Impact: Empowering Communities

By continuously refining our predictive model, we move closer to a future where water shortages in rural Tanzania are drastically reduced. This proactive approach ensures consistent access to clean water, empowering communities and fostering sustainable development.